

Ryan Cho

Senior Software Engineer | Backend, Data & AI Systems

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PROFESSIONAL SUMMARY

Software engineer with 8+ years delivering production backend systems and data pipelines across finance and SaaS. At Aspen Capital, built Python services with FastAPI and Celery, shipped LLM-powered retrieval and agent workflows with guardrails and evaluation suites, and hardened observability with traces, alerts, and SLO-driven rollback patterns. Strong in SQL schema design, RAG pipelines, hybrid search, and cross-functional delivery with Product and Design.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected **production LLM agent services** in Python and FastAPI with tool calling, multi-step planning, and function orchestration, enabling research and document drafting workflows adopted across internal investment teams.
- Built **retrieval and evaluation pipelines** with embedding generation, chunking strategies, hybrid keyword-vector search, and cross-encoder re-ranking, improving grounded answer quality within defined latency and cost budgets.
- Established **guardrails and automated eval suites** using Ragas and DeepEval for RAG, extraction, and summarization tasks, combining human-in-the-loop review with CI gates that blocked quality regressions before production deploys.
- Hardened asynchronous processing with **Celery workers and Redis**, designing idempotent tasks, retry policies, and fallback paths that maintained SLAs during peak batch and interactive request load.
- Owned **observability and incident response** for backend and data services—structured logging, distributed traces, dashboards, and alert runbooks that shortened detection time for production anomalies.
- Partnered with Product and Design to scope AI features, define success metrics, and run **A/B and canary experiments**, translating experiment outcomes into iterative model, API, and UX improvements.
- Raised engineering standards through **comprehensive test coverage**, API documentation, and thoughtful code review, mentoring engineers on SQL schema design, query tuning, and resilient service patterns.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Python backend APIs and data services** for trading and risk platforms, shipping features from prototype to production with unit, integration, and contract tests.
- Designed **PostgreSQL schemas and SQL queries** supporting high-volume transactional reporting, optimizing indexes and query plans that shortened batch processing windows for downstream analytics.
- Implemented **Dockerized microservices on AWS** with CI/CD pipelines, enabling repeatable deployments and rollback strategies for critical market-hours systems.
- Built **ETL pipelines** ingesting market and reference data with validation rules and monitoring, improving data quality for compliance-facing reports and internal dashboards.
- Collaborated with business and engineering stakeholders to translate requirements into **technical specifications**, delivering reliable integrations under tight regulatory deadlines.
- Contributed to **observability practices**—centralized logs, metrics dashboards, and on-call playbooks—for distributed services supporting global banking operations.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Shipped **Node.js and TypeScript features** for a SaaS inspection platform, integrating REST APIs and improving form workflows used by field operators.

- Wrote **unit and integration tests** for backend endpoints and data validation logic, increasing release confidence across bi-weekly deploy cycles.
- Assisted in **PostgreSQL data modeling** for customer and audit records, supporting reporting features with normalized schemas and efficient queries.
- Participated in code reviews and Agile ceremonies, applying **production deployment practices** including Docker containers and staged rollouts.
- Debugged production issues with structured logging, reducing recurrence of **customer-facing defects** in payment and notification flows.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web interfaces** with JavaScript and CSS for a nonprofit foundation site, improving accessibility and mobile usability for donor-facing pages.
- Integrated **backend form handlers and content APIs**, enabling staff to publish updates without developer intervention.
- Collaborated with designers to implement **UI components** aligned with brand guidelines while maintaining cross-browser compatibility.
- Delivered features on tight deadlines during a **three-month engagement**, documenting handoff materials for ongoing maintenance.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: TypeScript, JavaScript, React, HTML5, CSS3, Node.js

Backend: Python, FastAPI, Celery, Ruby on Rails, LangChain, LangGraph, LlamaIndex, REST APIs

Databases: PostgreSQL, SQL, Redis, Schema Design, Query Optimization, Vector Stores

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Terraform

Monitoring & Tools: OpenTelemetry, Datadog, Grafana, Prometheus, Ragas, DeepEval, OpenAI Evals

Workflow & PM: A/B Testing, Canary Deployments, Incident Response, Agile, Cross-functional Collaboration